

CONTROL AND INSTRUMENTATION OF THE PROTOTYPE FAST REACTOR

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1. Introduction

PFR poses some new problems in control and instrumentation. This is an early stage of the project and current development work will influence the final choice of instrumentation to meet these problems. Therefore, in discussing the principles of the proposed control and instrumentation systems, the paper lays stress on the problem areas and the development work being undertaken to provide a solution.

The PFR essentially consists of a reactor core cooled by a primary sodium circuit and a secondary sodium circuit transferring its heat to the steam-raising plant and turbine. The primary and secondary circuits are coupled together through an intermediate heat exchanger. The plant is designed to deliver 250 MW(e) into the grid system. The control system must therefore match the reactor power to the load demand via the secondary circuit, delivering the steam at all times at the correct conditions for turbine operation. It is also desirable that load fluctuations should, as far as possible, not be reflected into the reactor as control rod movements. The system finally adopted is discussed in the section on plant control.

The central instrumentation problem in any reactor lies in ensuring the correct operation of the core. Not only does it have to be operated to provide the necessary steam conditions, but also it must be operated safely, that is to its design conditions. Departure from design conditions must be corrected and, in the event of a positive fault, the reactor shut down. This is the subject discussed in the section on core monitoring. For correct operation the core must have a guaranteed supply of sodium coolant at the correct conditions. The instrumentation to meet this requirement is discussed in the section on sodium instrumentation. Because of the technical similarity of the instrumental problems in the primary and secondary circuits, the requirements for

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the secondary circuit are also discussed in the same section. The possibility of a sodium/water reaction poses a special problem in the secondary circuit, but its discussion is considered more appropriate to another paper to this conference.⁽¹⁾ Finally, the handling and presentation of the information from these instrumentation systems is described.

2. Control System

The design requirement is that the PFR should fit into the grid system and must therefore be able to follow the changes of electrical load demanded

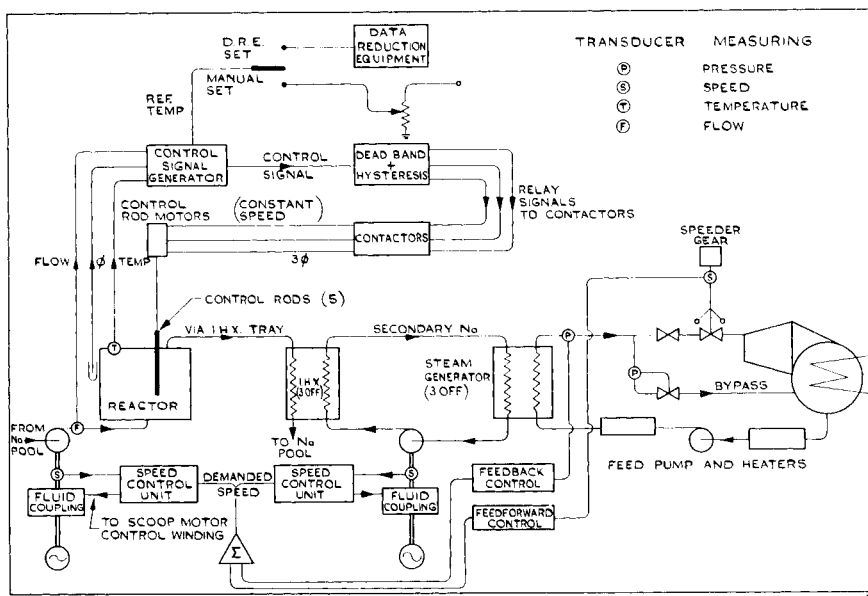


FIG. 1. Automatic control system.

by the system. It is also desirable that unscheduled load changes of small amplitude are smoothed out to avoid excessive movement of control rods.

To achieve load following, two quantities have to be controlled. First, the power generated in the core has to be matched to the load demand. Secondly, the operating conditions of the primary and secondary circuits have to be set to transport the required amount of heat from the core to meet the steam requirements of the turbine. This is accomplished by two closed control loops. One controls reactor outlet temperature by movement of control rods, the set point of this system being adjusted as a function of load so as to maintain constant steam temperature. The other keeps the steam pressure constant by a feedback of error in steam pressure to sodium flow. Figure 1 is a schematic diagram of the control system.

The reactor power is adjusted by movement of the control rods. When the

grid load increases, the primary circuit flow will be increased to match the new conditions; if no further changes were made to the system, the core outlet temperature would drop. Accordingly, the power of the core is adjusted to maintain the coolant outlet temperature at a set value. At the higher loads, larger temperature differentials are necessary in the plant to transport the heat across the heat exchangers from the core to the steam generator. Thus, the set temperature for coolant outlet from the core is not constant but increases with load. The reference temperature for each particular load demand is calculated according to the state of the plant by the data reduction equipment.

This reference temperature is compared with the core outlet temperature, which is taken from three thermocouples on the outlet of three subassemblies located near the centre of the core and in such a position that their ratings are not appreciably influenced by local flux changes from movement of control rods. A control signal generated only by the difference between the core outlet temperature and the reference temperature would lead to an undesirable overshoot of the power. To stabilize the control system, the control signal includes a term based on the rate of change of power derived from a flux measurement. That is, the control signal is of the form

$$(T_{\text{ref}} - T) - a \frac{d\phi}{dt}.$$

The term a is not a constant and its optimum value increases as the primary circuit flow decreases. To take account of this, the control signal is further modified by incorporating a flow term in the form

$$(T_{\text{ref}} - T) - k \frac{d}{dt} \left(\frac{\phi}{F} \right).$$

The quantity k is now a true constant, and this form of control signal has the advantage that when the load is being changed the time-dependent term disappears (primary circuit flow is nearly directly proportional to power).

The control rods are actuated by an ON/OFF device. There is a dead-band the width of which is set by the noise in the reactor and the stability of the control system. Only $\pm 1^\circ\text{C}$ dead-band is required for stability; a greater width may be required if the reactor noise level proves to be higher than expected. Hysteresis is also incorporated in the system; it allows some continued movement of the control rod after the error signal enters the dead-band, and thus eliminates unnecessary small movements of the control rod. The system will operate on one or more control rods, the number in use at any time depending on the reactivity worth per unit movement at the position of insertion. The operator can ensure that all control rods are maintained at about the same level, so that asymmetries in flux due to non-uniform insertion of control rods are avoided.

The second closed control loop maintains the outlet steam from the steam generator at constant pressure by adjusting the flows in both primary and secondary circuits, which is done by varying the operating speed of the pumps. When there is a change in the load demand of the system, there will be a time lag before the pressure falls outside the dead-band of the controller, and thus before the sodium flows are adjusted to the new load conditions. To eliminate this time lag for predetermined load changes, a signal from the turbine speeder gear is fed directly to the pump speed control units, so that on resetting the turbine load, the pump speeds are adjusted to give the appropriate sodium flows without waiting for this to be established through the pressure controller. The steam generator has a large thermal capacity, and advantage is taken of this in conjunction with the dead-band on the pressure controller to ensure that small unscheduled fluctuations in the load are absorbed at this point and not reflected back into the reactor circuit.

At low power operation, steam from the generator by-passes the turbine directly to the condensers. At start-up, the by-pass flow will be progressively closed as the turbine demand increases. To facilitate this operation, the steam by-pass is controlled by a proportional control using feedback of the steam pressure in the system. The set pressure is slightly higher than the set pressure in the main control system which operates on the primary and secondary flows. Hence these flows are kept on their lower limit, when the by-pass is operating. When the turbine demand increases beyond the point where the by-pass closes completely, the pressure falls and control passes automatically to the main control system.

The stability of the plant and the performance of the various control arrangements have been studied extensively, using analogue and digital computers. Simulation of the primary circuit on a PACE computer showed the reactor to be stable over the expected range of parameters.⁽²⁾ Indeed, when values of the Doppler and the coolant coefficients well outside the predicted range were used in the analysis, the plant remained dynamically stable. Studies on a dynamic model of the complete PFR plant showed that the whole plant is extremely stable and will absorb load changes without difficulty.

In addition to conventional measurements of pump speed and steam pressure, the control system requires temperature and flux in the core to be measured. The special problems posed by the PFR core for flux and temperature measurement are discussed in the following section on core monitoring.

3. Core Monitoring

Core monitoring instrumentation is required to show that the power input and heat removal systems are operating to the required design conditions, and to detect any departure from these conditions. The fundamental design

criterion is that the fuel-pin clad temperature should be below a maximum hotspot temperature, which is chosen to give an insignificant probability of cladding failure.⁽³⁾ A basic tenet of the safety philosophy is that damage to the core can only occur if there is a significant fuel pin failure;⁽⁴⁾ hence it is of vital importance to operate to the required design conditions and to detect any departure from them. It is physically impracticable to measure the cladding temperature directly in a fast reactor, where the core must be

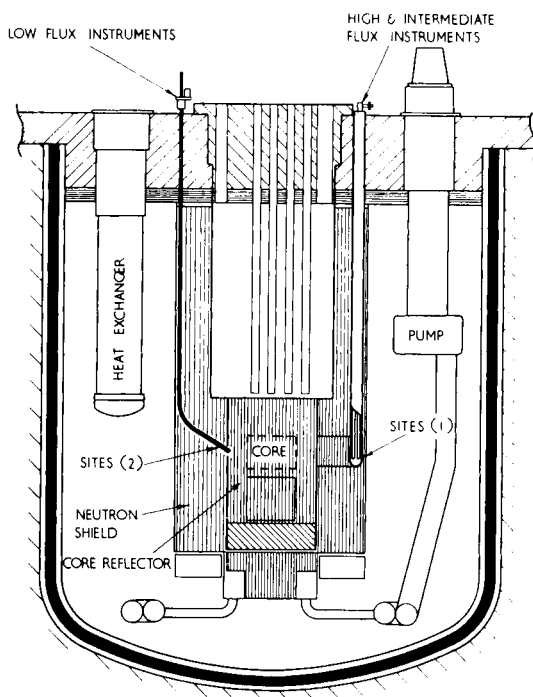


FIG. 2. Section showing detector layout.

close packed to give high power densities for economic operation. The operating conditions, therefore, have to be deduced from the power input to the core by flux monitoring; and the outlet temperature of the subassemblies, together with the knowledge of the total core flow and the flow calibration of the individual subassemblies.

4. Flux Measurement

In PFR, flux measurements have to be made within the primary containment because the integral design of the reactor makes the containment boundary too remote and well shielded from the core. Access has to be from

the top to avoid penetration of the reactor vessel, but the area of the rotating shield has to be avoided. Further, instruments must be easily replaceable for maintenance. Suitable positions to meet these conditions for intermediate and high power measurements are found in the side wall of the neutron shield. Figure 2 shows the flux detector sites, which demand instruments and cables which can operate in the temperature range 560–600°C. This, combined with large gamma background due to the sodium coolant, poses

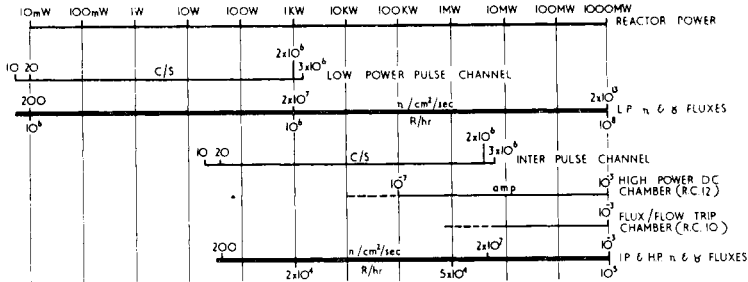


FIG. 3. Power coverage by nucleonic instruments.

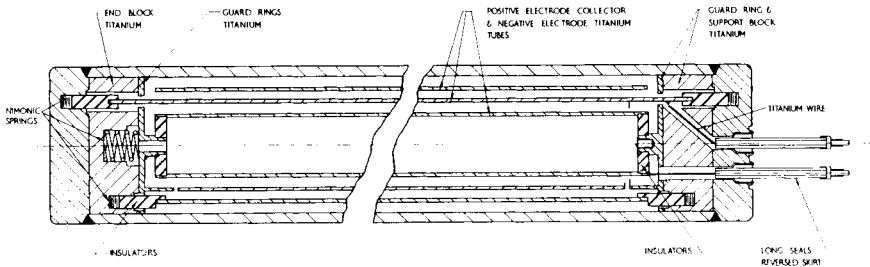


FIG. 4. RC12 compensated ionization chamber.

difficult instrumental design problems. Operating neutron flux and gamma backgrounds at these chosen sites are illustrated in Fig. 3.

Instruments indicating linear power, rate of change or deviation of power, and excess flux trips need cover only two decades in the power range. Gamma compensation is unnecessary, and an ionization chamber, the RC10,⁽⁵⁾ which can operate at 400°C with some degree of cooling could perform this function. Alternatively, a new 550°C design employing titanium electrodes,^(6,7) known as the RC14, could be used with little or no cooling. Site 1 (Fig. 2) is suitable for both designs. Wide-range logarithmic measurements require gamma compensation, and a 550°C compensated chamber, RC12 (Fig. 4), is being developed to operate at Site 1. It will work down to

100 kW shortly after operation at full power. For this performance the chamber sensitivity is high, delivering 10^{-7} A at 100 kW. Using recently developed cables,^(8,9) which have leakage currents below 10^{-9} A at 550°C, measurement down to a few kilowatts is possible when gamma levels have been allowed to decay. Thermionic logarithmic elements cannot accept the full power currents from the chambers (10^{-3} A or more) but a compatible transistor logarithmic element is available covering the range 10^{-10} – 10^{-2} A.⁽¹⁰⁾

To ensure an adequate reactivity shutdown margin the reactor shutdown power is about 20 mW. The pulse channels therefore have to cover a range of about $3 \times 10^8:1$ to provide overlap, and this range cannot be covered by a single logarithmic channel. However, a fast-pulse channel with a range of 10 – 3×10^6 c/sec has been developed;⁽¹¹⁾ it can handle pulses of 20 ns, and two channels will adequately meet these requirements. For these channels a fast-collection, high-temperature detector is also available, which will be used in Site 2 in Fig. 2 and will not be cooled.⁽¹²⁾ Since the chamber can operate overloaded, at about 10^{12} n/cm² sec, very little retraction is necessary at full power.⁽¹³⁾

Although existing techniques are adequate to meet the PFR requirements, developments⁽¹⁴⁾ could well produce improved instruments before the date for the final specification for PFR.

5. Temperature Measurement

Temperature measurements in sodium present no serious problems. Mineral-insulated chromel/alumel thermocouples sheathed in stainless steel are rugged and corrosion resistant, and have adequate sensitivity and response times. The operating conditions of most subassemblies will be obtained by mounting this type of thermocouple at the subassembly outlet. There is, however, a problem in providing a representative mixed-mean sample at the thermocouple. Adequate mixing will not ensue from turbulence in the exit portion of the subassembly unless the length is extended by some 12 ft (4 m). Two devices are now available which achieve the same mixing within the space available at the exit of the subassembly for a pressure drop of a few lb/in.² (0.2–0.3 atm). The first uses a number of tubes to obtain samples from each region, which are mixed through an orifice to give a final sample.⁽¹⁵⁾ The second simply rotates the fluid, and sufficient mixing is obtained over the central axial region two or three diameters downstream.

6. Fault Detection

Departures from design power input conditions are detected by conventional flux monitoring techniques. Maximum reactivity steps are limited

by design, and rates of addition are restricted in the case of maloperation to give acceptable temperature transients; trips are provided for period, temperature and flux/flux ratio.

Departures from design operating conditions in the heat removal system are extremely difficult to detect because the average flow and temperature at subassembly outlet are particularly insensitive to local high-temperature conditions (local blockages), due to the finely divided nature of the core and the large pressure drop across it. However, such local blockages could lead to dangerous situations in the core, and therefore instrumentation is required to detect a fault condition as soon as possible after it occurs and certainly, if it can propagate, before any damage can spread outside the confines of a single subassembly. Theoretical studies of pessimistic upperbound models⁽⁴⁾ indicate that the propagation rates could be fast. This has led to the study of high-sensitivity techniques as an early warning of fault conditions, before damage (clad failure) can occur, and of fast-acting systems, which can shut the reactor down in the event of clad failure, or, in the extreme, limit local meltdown. If such faults are detected, the failed subassembly must be located, and this means that samples must be taken from individual subassemblies. The same instrumental principle must be used for location as for detection, to avoid ambiguity. In principle, therefore, burst cartridge detection systems will be installed in PFR not only to detect, but also to locate, failure in individual subassemblies. A rate-of-change-of-reactivity instrument is proposed to detect extreme events. It has not yet been possible to establish an early warning system, which could operate before cladding failure.

Two burst cartridge detection systems are being studied. The first detects radioactive fission products which find their way into the coolant stream, at least by direct recoil, in the event of fuel being directly exposed by cladding failure. The second is based on the fact that the pins are designed to retain fission product gas released from the fuel; thus a significant pressure builds up in the pin, as a function of burnup, and will be released on pin failure.

The problem of using radioactive methods in a sodium cooled reactor is to distinguish the activity of the fission fragments from that of sodium-24. This may be done either by stripping the radioactive gases from the liquid or by monitoring for the neutrons from the fission fragments.⁽¹⁶⁾ The second method is preferred and has been studied in detail. The monitor is expected to detect a few square centimetres of exposed fuel provided that (a) the fissile contamination of the surface of the fuel pins is closely controlled, and (b) the sodium is supplied clean and kept clean even after a fuel burst. The time response is dominated by the transit time from the fault to the monitor: it is expected to lie within the range 10–30 sec, the lower value being limited by the hydraulic design of the reactor.

The gas released from sudden failure of a fuel pin can be collected into

a vortex by a cyclone placed at the subassembly exit,⁽¹⁷⁻¹⁹⁾ and detected by temperature, pressure or acoustic techniques. The sensitivity of these is being assessed.

Voids in the coolant of a subassembly may increase or reduce reactivity, according to their positions. Judd⁽²⁰⁾ has calculated that the reactivity change cannot be expected to be more than a few cents. Similar calculations show that the fuel melting and subsequent rearrangement could give reactivity changes of the order of half a dollar. Rapid changes of reactivity cause an almost immediate change in reactor power level, and it should be possible to set a trip on a high power flux monitor to be actuated by a change of a few per cent in power level caused by undemanded change in reactivity, i.e. not by a control rod movement. This is equivalent to several cents change in reactivity, so that only fuel meltdown would be detected. But, nevertheless, such a protection is an ultimate safeguard for this extreme accident. The precision with which the trip levels can be set will depend on reactor noise level and normal power changes.

Two systems which could give early warning of a fault condition, before cladding failure, are being studied. The first system examines the disturbance of temperature following a local blockage; the second assumes that as the fault condition propagates temperatures will reach the boiling point of sodium and that the boiling can be detected acoustically.

Although the measurement of average subassembly outlet temperature is too insensitive to detect a local blockage, a new principle⁽²¹⁾ of outlet temperature monitoring has been investigated quantitatively. It depends on the fact that a local blockage, in present designs of fuel cluster, produces a corresponding change or blip in the radial temperature profile in the sodium which persists into the exit of the subassembly, although with some dispersion. It may be detected at an appropriate position downstream by measuring statistically the strength of the fluctuations of temperature arising from the turbulent mixing of the hotter blip with its colder surroundings. In general, this measurement of root-mean-square temperature (in a selected band width) is about ten times more sensitive in detecting blockages than the more conventional measurement of mixed-mean temperature.

In the studies on detection of boiling, early experiments^(22, 23) showed that the acoustic spectrum of sub-cooled boiling water was broad-band, probably arising from the Rayleigh collapse of vapour bubbles in sub-cooled liquid giving significant high frequencies in the spectrum. Experiments in both water and sodium, in tubular and cluster arrangements, have been undertaken to establish a quantitative relation between the heat flux and acoustic power. As boiling is not expected to give a characterizable spectrum, it has to be detected against a similar spectrum from background noise in the reactor, the main source of which will probably be the pumps. Spectral measurements have therefore been taken on the 6000 gal/min (0.452 m³/s)

sodium pump⁽¹⁾ which is typical of PFR designs. It is not yet clear whether the signal-to-noise ratio will be adequate.

The acoustic signals can be detected in the reactor either directly by a capacitance transducer, or by piping the signal along a waveguide to outside the reactor vessel where standard piezoelectric transducers can be used.

7. Sodium Instrumentation

The reactor layout and the high temperature (600°C) of the sodium has led to the development of new instruments to meet specific requirements of flow and level measurement. Impurity in the sodium must be controlled below a specified level to prevent blockage and limit corrosion. Sampling followed by chemical analysis is the most reliable method of identifying and measuring impurities in sodium, and sampling systems will be provided on PFR. However, in-line impurity monitors are required for continuous operational control. Both rhometers⁽²⁴⁾ and plugging meters are proposed. Recent developments in plugging meters are discussed below.

8. Flow Measurement

Primary Circuit Flowmeter (Transit Time Flowmeter)

The integral design of the PFR, with the three primary circuits immersed in a sodium pool, together with its compactness, precludes long pipe runs and makes the use of conventional flow-measuring techniques difficult. A new flow-measuring technique is therefore being developed to meet these conditions.

The flowmeter⁽²⁵⁾ is based on the measurement of the transport time of a thermal noise pattern between two thermocouples spaced a known distance apart, from which the fluid velocity, and hence mass flow, can be deduced. The thermal noise is inherent in the primary circuit and is due to temperature fluctuations generated by fluctuations in heat-transfer conditions. The thermocouples need only be spaced a few diameters apart. The thermocouple signals (at the microvolt level) may be amplified, recorded on tape and subsequently analysed by filtering and phase-angle measurement. However, on the PFR, where a computer is available, the measurement can be done directly using continuous real time correlation.

One advantage of the method is that the sensors (thermocouples) are small, reliable, easily duplicated and replaceable without great complication. Another is that the transport time is substantially independent of flow distortion in the duct, as predicted by Taylor.⁽²⁶⁾ The disadvantage is that long measuring times (100–1000 sec) are necessary.

Secondary Circuit Flowmeter

The secondary circuit has easily accessible pipework and long pipe runs so that conventional electromagnetic flowmeters can be used. These give the advantage of continuous and more precise measurement than is possible with the transit time flowmeter in the primary circuit. With a 23 in. (57.5 cm) diameter pipe a very large, clumsy magnet would be necessary if a conventional electromagnetic flowmeter was used. This has led to the development of a new form of saddle coil flowmeter using current excitation.

The flowmeter works on the orthodox electromagnetic flowmeter principle, but the magnetic field is supplied by windings energized with direct

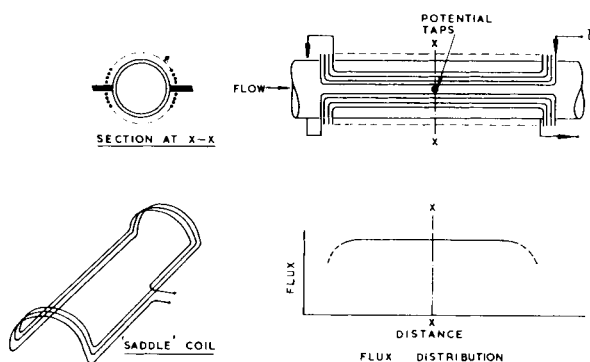


FIG. 5. Saddle coil flowmeter.

current. These are made in rectangular spiral form, curved to conform to a cylindrical surface somewhat greater in diameter than that of the measured duct (illustrated in Fig. 5). Two such windings are applied to opposite sides of the flow ducts. Coil pairs of this "saddle" form generate a very nearly homogeneous field within the enclosed duct volume, the reproducibility of which is a function of the stability of the system geometry and of the energizing current. From this uniform field distribution (which can be either calculated or measured on a scale model), and the dimensions of the system, the calibration and corrections can be calculated to better than $\pm 5\%$ of flow.

The first experimental saddle-coil flowmeter for a 12 in. (30 cm) diameter duct used two coils each of 25 turns of aluminium conductor, aluminium sheathed, mineral insulated cable. It gave a field falling by 10% peripherally over a length equivalent to $2\frac{1}{2}$ diameters. A comparison between this and the transit time flowmeter on a flow of 5000 gal/min ($0.375 \text{ m}^3/\text{s}$) gave a maximum difference of less than 10% which was within the expected accuracy of this experiment.

Flow Failure Monitor

Loss of coolant is made incredible by design. However, coolant flow to the core could be lost by failure of either the pump or the inlet duct to the diagrid. The safety considerations are discussed in a separate paper to this conference.⁽⁴⁾ Fast acting flow monitors are specified as flow failure alarms on all pumps. The pump monitors will not give a unique signal in the event of duct failure to the diagrid. To guard against this particular fault the flow through the core must be measured directly. This will be done by mounting a flow failure monitor on each of three dummy subassemblies located at the periphery of the core, set to trip on a rate of change of flow.

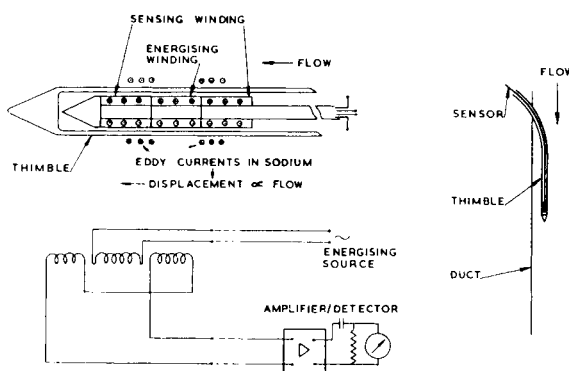


FIG. 6. Pump flow failure monitor.

The flow monitor to detect pump failure is inserted into a thimble projecting into the measuring station downstream of each pump. Its operating principle is essentially that of the Lehde and Lang flowmeter.⁽²⁷⁾ The sensor⁽²⁸⁾ consists of a cylindrical head containing a central winding energized from an a.c. supply, flanked on either side by two similar windings, connected differentially so that their net output is nominally zero with no sodium flow (Fig. 6). With sodium flow, the motion of the liquid metal in the magnetic field of the energizing winding induces asymmetric eddy-currents in the sodium, which unbalance the e.m.f.'s induced in the sensing windings: in effect, the magnetic field is displaced downstream by the liquid metal flow. The resulting output signal is approximately linearly related to local sodium velocity. The velocity calibration has no simple relationship to the total duct flow. For alarm purposes, the sensors will be operated on a relative output basis, but they can be calibrated *in situ* in terms of duct mass flow from values derived from the primary and secondary flowmeters.

The duct failure monitor is similar in design, but will be directly immersed in sodium and located at the outlet of a dummy subassembly, the flow in

which is representative of total core flow. It can be replaced at shutdown via the fuel charge/discharge route.

9. Level Measurement

The hydraulics of the PFR with its free surfaces, require the setting and control of a number of sodium levels, the sodium having a maximum temperature of 600°C . One type of sensor is being developed to meet all these

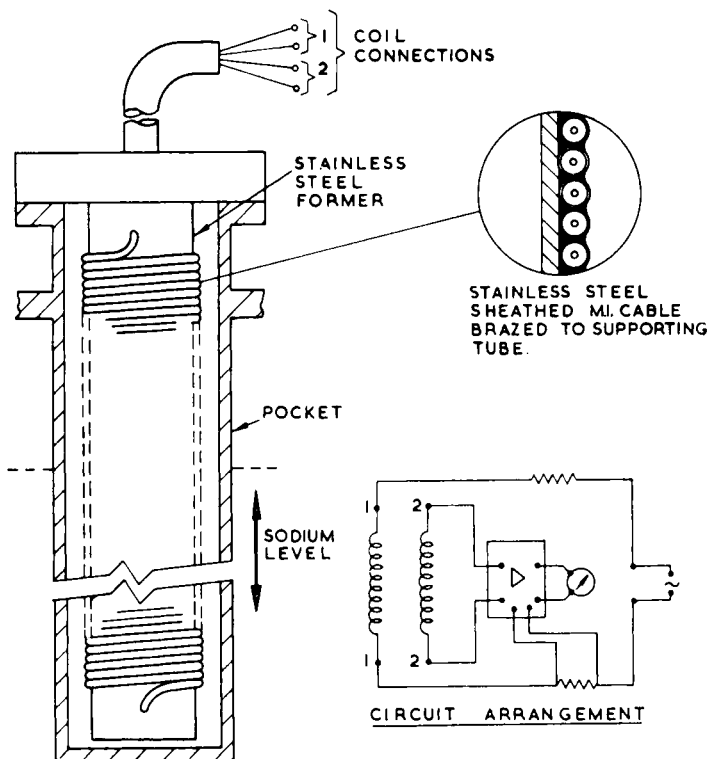


FIG. 7. Sodium level sensor.

requirements. It operates on the differential transformer principle where the shorted-turn effect of liquid sodium varies the inductive parameters of a double-wound inductance. The windings are stainless steel sheathed, mineral insulated cables of coaxial form, brazed to a central support tube⁽²⁹⁾ (Fig. 7). This sensor is mounted in a thimble which projects into the measured vessel. Immersion by rising sodium reduces the mutual inductance (typically a few microhenries) linearly with increasing depth, apart from end effects. The power sensitivity of the system is low, but this is offset by the fact that cables,

using Nichrome conductors, have excellent high temperature characteristics so that relatively high energizing powers can be used if required.

10. Impurity Monitoring

An experimental design⁽³⁰⁾ of continuous automatic plugging meter is shown in Fig. 8. In operation, the sample flow divides between the fixed flow restrictor and the plugging annulus. On cooling below the impurity saturation temperature, the annulus begins to plug. The control system seeks to maintain the annulus flow at a small value, equivalent to the partially

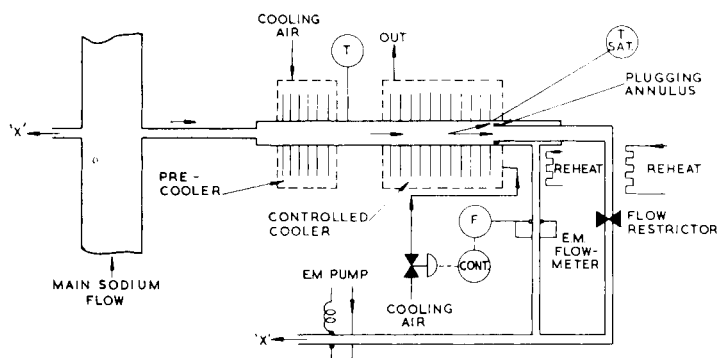


FIG. 8. Automatic plugging meter.

plugged condition, by adjusting the flow of cooling air. At the controlled equilibrium (i.e. with the establishment of the partially plugged condition) the annulus temperature coincides with the saturation temperature which is measured and indicated as the output signal. The constant head auxiliary pump avoids the need to measure flow ratio and also allows the meter to be independent of flow changes in the sampled circuit. The flow sharing principle is adopted, not only to give good sensitivity for a given mass of precipitated impurity, but to provide an adjacent continuous flow into which an accidental total plug of the annulus can redissolve reasonably easily.

The meter has operated satisfactorily with indicated plugging temperatures from 250°C to 125°C. The relation between saturation temperatures and impurity concentration is uncertain,⁽³¹⁾ but concentrations down to the equivalent of about 6 ppm of oxide can be followed; these are well below the nominal working level of 10 ppm.

11. Data Handling and Presentation

The handling of the data gathered by the measuring systems discussed above (from some 2500 measuring points) and their presentation to the plant operators constitute a major problem of system design. Fully computerized

control of the plant was considered, but was rejected as representing a complication to a plant whose prime purpose is to demonstrate the reactor system. The proposed system is therefore designed on the following principles:

- (i) Major plant items will be started up from local control points, i.e. about ten locations.
- (ii) After start-up all subsequent control, including shutdown, will be directed by either the central control room (CCR) staff or automatic control equipment.

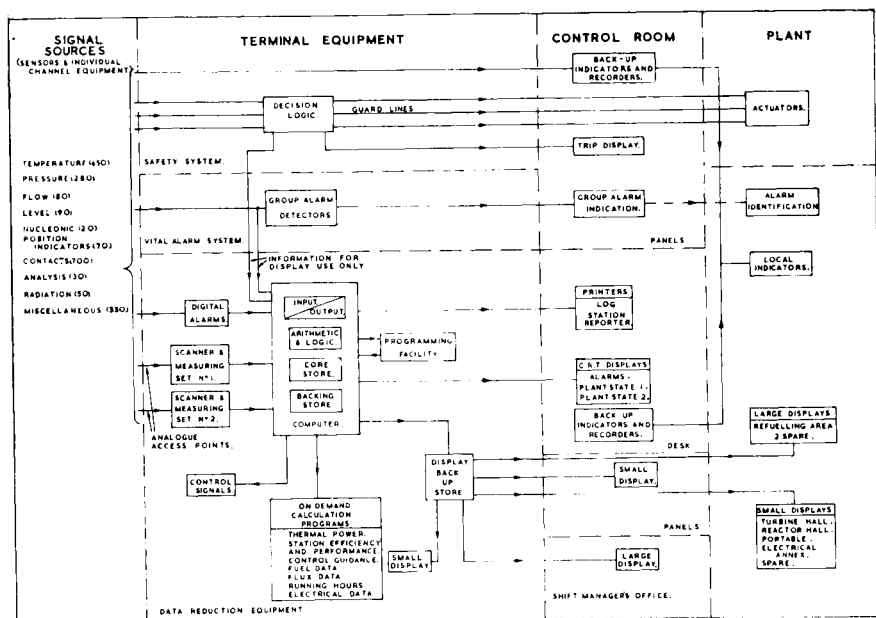


FIG. 9. Data handling and presentation.

- (iii) Automatic data reduction equipment (DRE) will form the basis of all information processing and display in the CCR.
- (iv) The DRE must be available for start-up, but the plant should never have to shut down because of a DRE failure. There will be adequate back-up instrumentation for steady state running.
- (v) All plant alarms will be surveyed and displayed by the DRE. Back-up alarms will be available in the event of DRE failure (the vital alarm system).
- (vi) The reactor safety and protective interlock systems for plant items should act independently of other instrumentation.

Figure 9 is a schematic diagram of the overall arrangements to meet these needs. The DRE, together with the separate alarm and trip facilities, will

give all the information required to operate the plant, and the necessary controls to do this will also be located in the CCR.

The DRE comprises an input multiplex switch or scanner, analogue-to-digital converters, a digital computer, cathode ray tubes for output data display and electrical printers.⁽³²⁾ The overall accuracy of measurement from system input terminals to display will be better than 0.5%. System availability should be better than 99.5% with a minimum "mean time" between system failures of 2000 hours.

As the DRE is computer based, a stored programme is used to order all operations required to organize, compute, display or store the incoming data. In the PFR, therefore, the novel problems lie in the design of the input and display systems.

There are 1000 digital inputs which go directly into either the computer store or a separate alarm memory. Twenty per cent of these inputs will be tested for change of alarm state every 100 msec, and the remainder every 5 sec. There are also 1600 analogue signals which have to be handled at a rate of 200 inputs/sec. The scanner is divided into two main sections each with its own analogue-to-digital converter. This arrangement effectively halves the required input rate and increases reliability, which can be further improved by connecting inputs from the same functional group to different sections of the scanner. Input filters may be required to reduce noise. In this case, the cost of filtering can be reduced by using two stages of multiplexing in the scanner and interposing filters between them so that, on each scan, several inputs share a filter.

Random access is not mandatory for the great majority of inputs, where operational needs can be met by sequential scanning at an overall rate of 200 inputs/sec. Hence, sequential rotary scanning switches are being studied because of their potential simplicity and cheapness. Of the two types being investigated, the first operates by passing a permanent magnet on a rotary arm over a ring of dry reed elements mounted on a Dural stator, with Swedish iron insets providing a magnetic circuit to fix the ON/OFF time of individual reeds. Switching rates of 100 connexions/sec have been reached with satisfactory mark-space ratio. In the second switch, the contacts are metal studs heavily plated with soft gold and embedded in a plastic of similar hardness to the plating. These are wiped by a brush of rather harder gold alloy. Tests have reached the equivalent of 100 connexions/sec for 1 year, but this is not adequate information to estimate life and reliability. The wiped contact switch, unlike the reed, exhibits no internally generated noise transients and can be used at higher switching speeds.

The information required for plant operation will be displayed on cathode ray tubes, chosen (a) as the best way to give information to the operator, and (b) because they simplify the layout of the control room. The display will consist of regularly updated alpha numerical data, either tabulated or super-

imposed in mimic diagrams, as well as the graphical presentation of selected past data. Electrical printers are required for logging statutory data and for permanent record.

12. Conclusions

Study of the control of PFR has revealed no serious problems, and the reactor is expected to be very stable under all conditions.

Based on existing knowledge, solutions are available for problems of core flux and temperature monitoring and of sodium instrumentation. Current development work should yield significant improvements in these designs, and may lead to new proposals.

In principle, a burst cartridge detection and reactivity monitor will be provided for detecting subassembly faults. However, the problem of fast detection of local blockages in a subassembly remains to be solved.

The philosophy of local control points for reactor start-up, with all the necessary controls in a central control room for subsequent reactor automatic power operation, is seen as a step towards full plant automation. The proposed computer-based data reduction equipment allows simplification of the control room and, together with the use of cathode ray tube displays, gives a good ergonomic design.

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